GTC4Lusso USA - Checking service brakes for wear - Latest Update: 2021-01-26

A3.21 Checking service brakes for wear

Checking service brakes for wear

➤ Remove the complete wheels (D2.01).

- At the intervals indicated in the Maintenance Schedule, check the state of wear of the brake pads and discs.
- The front brake pads are fitted with a wear indicator connected to the brake failure warning light on the instrument panel. Should this warning light illuminate, check the thickness of the pads and the state of the braking surfaces on the discs.
- In the event of excessive wear, even of just one pad, replace all the pads on the axle.

➤ Refit the complete wheels (D2.01).

Checking and inspecting the brake discs

The brake discs must be perfectly clean, free of any oil, grease or rust, and must not be excessively scored.

In the event of poor braking performance not attributable to the state of the brake pads and callipers, carry out a series of dimensional checks on the discs.

- Using a dial gauge, measure the planarity with the disc fitted on the wheel hub.
- Using a micrometer, measure the parallelism between the contact surfaces and the pads at different points on the braking surface.
- Using the special tool, check the roughness of the pad contact surface.

➤ The rear discs differ from the front discs in that they have two additional holes to enable access to the brake shoe adjustment pin without removing the disc.
➤ They also have a large friction surface on the inner side for the parking brake.

Checking CCM brake disc wear

- This operation includes:
- visually inspecting for alterations in the brake disc friction surface;
- checking the minimum brake disc thickness (maximum permitted wear **0.5 mm**);
- checking the brake disc edges for damage (damage caused by improper use).

- The brake discs must be replaced if:
- there are significant alterations in the friction surface;
- the brake discs are worn beyond the minimum required thickness (material abraded).

- Normally, both forms of brake disc wear occur simultaneously. Only rarely, following prolonged stress on the race track with large temperature variations on the friction surface, surface alteration may become so pronounced as to necessitate earlier replacement of the brake discs.

Always replace the CCM brake discs on both sides of the same axle.

- Front and rear CCM disc wear is calculated by the instrument panel operating software, which uses an algorithm to determine when to switch on the relative indicator to warn the driver.

The bedding-in procedure must always be performed after replacing the brake discs or the brake pads.

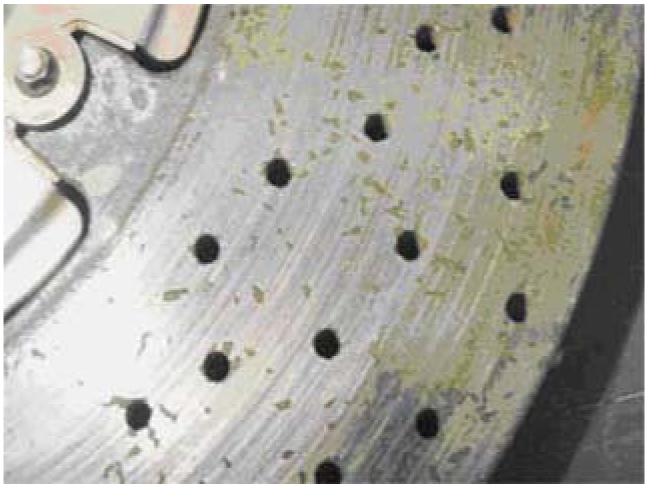
- Disc replacement is necessary when the wear limit is reached, as indicated by the relative warning on the TFT display, or if visual inspection reveals any surface damage.

- **Visual inspection of the disc friction surface**

- The alterations in the friction surface are caused by the material oxidising as a result of extremely high thermal stress. The high temperatures reached by the brake discs, especially in race track conditions, cause the material to oxidise progressively.
- Surface alteration is detrimental to braking comfort, reduces brake disc resistance and increases pad wear, requiring the replacement of components earlier than scheduled.

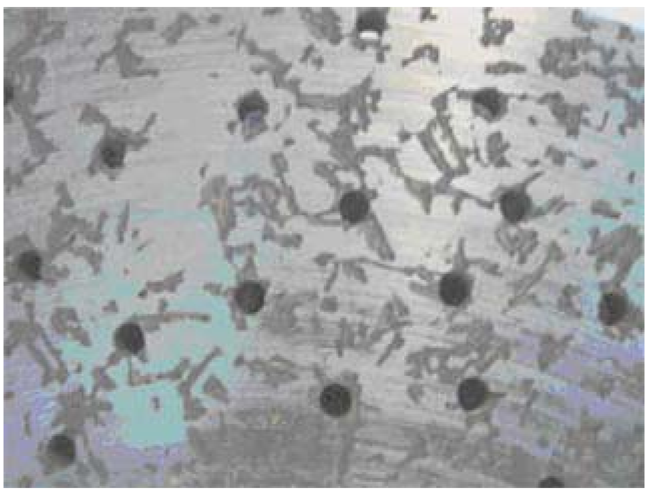
- Understanding the signs of brake disc wear and consequent operations necessary:

- The figure aside shows the friction surface after normal brake usage in road conditions; disc replacement is not necessary in this case.



- The figure aside shows the friction surface after a long journey, with only one intense brake application; here too, disc replacement is not necessary.

- The figure aside shows the friction surface after an appropriate mileage or following maximum stress in track conditions; disc replacement is necessary in this case.



- The figure aside shows the friction surface after an appropriate mileage or following maximum stress in track conditions; disc replacement is necessary in this case.

- **Checking minimum disc thickness**
- The minimum thickness of cross-drilled discs must always be measured at the base of the inner or outer friction surface track. Use a suitable micrometrical screw or a brake disc template.
- Maximum permissible wear is **0.5 mm**

- **Checking disc edges for damage**

Regardless of the overall state of wear, the disc must be replaced if it shows signs of damage around the edge.

Any unauthorised mechanical work on the holes in the surface of the disc may damage the brake disc itself.

If necessary, clean the holes in the disc brake and the friction surface of the rotor with a pressure cleaner.

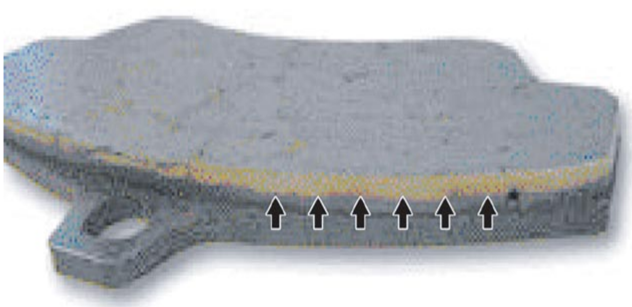
Do not perform any mechanical work on the holes in the disc.

CAUTION

When using pressure cleaners to clean discs, use adequate face protection.

- When assessing damage to the disc edges, apply the following criteria:
- maximum permissible width/depth **2 mm**;
- maximum permissible length **(10 mm)**;
- no more than **3** damaged areas are permissible.

Check CCM brake pad wear



After particularly intense use on the track, check the appearance of the pads as well as checking normally for wear. If there is a distinct white discolouration on the edges of the pads, which extends almost as far as the metal backing plate, the pads must be replaced to ensure that the brake system continues to perform correctly.

CCM brake disc wear

As the discs wear, their mass reduces. This is due to two simultaneous effects - a thermal effect, which burns the carbon fibres, and a mechanical abrasion effect between the disc and pad.

The instrument panel operating software uses an algorithm to calculate, in relation to effective vehicle usage (e.g.: deceleration caused by braking), separate wear indices for the front **P(a)** and rear **P(p)** carbon brake discs, expressed as a percentage (%) of wear.

These indices are periodically compared against a threshold value **(100%)**. Upon reaching this threshold, an amber warning signal (possible danger of disc failure) is shown on the TFT display, warning the user that disc replacement is necessary.

The user is warned of this condition by the appearance of the message: **“CCM brake discs worn”** on the display, together with the specific amber symbol. At the end of the display cycle, or when the MODE button is pressed (with **“Escape”** function), the symbol is reduced to icon form.

An RLI for “Number of brake disc replacements” is also generated, identifying the number of times that the discs have been replaced.

Once the brake discs have been replaced, the wear index may be reset on the instrument panel.

The wear index indicated on the instrument panel may only be reset when the discs are replaced, once the wear index exceeds 100%.

To maintain a record of the latest brake wear index reset, three values are stored in the statistics memory for the front and rear brakes, which are updated upon reset (with the possibility of storing the last seven resets):

- Number of brake disc changes counter: --------- (dimensionless number)
- Kilometres at first disc brake change: --------- Km
- Wear percentage at first disc brake change: --------- %

At each service, submit the following data (printing out the parameters from the instrument panel):

- vehicle serial number;
- vehicle mileage (“Odometer” parameter);
- “Disc Wear Index” value.

There are essentially **5** different scenarios possible:

- 1 - Replacing all carbon discs due to vehicle usage;
- 2 - Replacing one carbon disc (or two discs on same axle) due to failure;
- 3 - Replacing a working instrument panel (dialogue with diagnostic instrument possible);
- 4 - Replacing of a malfunctioning instrument panel (dialogue with diagnostic instrument not possible);
- 5 - Replacing the carbon discs before reaching wear threshold (“Disc Wear Index” parameter < 100%).

1 - Replacing the carbon discs due to vehicle usage

- In this situation, the amber warning signal on the TFT display is lit.
- Use the diagnostic instrument to verify from the instrument panel that the disc wear index has exceeded 100%.
- Replace the carbon discs.
- Reset the carbon disc wear index on the instrument panel with the diagnostic instrument.

2 - Replacing one carbon disc (or two discs on same axle) due to failure

- In this situation, only one disc (or two discs on the same axle) are replaced, without modifying the parameters on the instrument panel.
- In other terms, the “Disc Wear Index” parameter is left at the value for the discs not being replaced (partially worn).

3 - Replacing a working instrument panel (dialogue with diagnostic instrument possible)

Order a replacement instrument panel following the instructions given in **TI 2009** issued **May 2012**.

4 - Replacing a malfunctioning instrument panel (dialogue with diagnostic instrument not possible)

Order a replacement instrument panel following the instructions given in **TI 2009** issued **May 2012**, specifying:

- the serial number of the vehicle;
- estimated vehicle mileage.

SAT will verify the estimated vehicle mileage.

Using a table correlating vehicle usage time with increase in disc wear % compiled by the engineering division, SAT will determine the % disc wear value to be entered in the new instrument panel in relation to the mileage recorded at the last service.

Upon approval from SAT (which will verify the data supplied by the dealership), the Spare Parts Department will confirm the order and supply a spare instrument panel (with the estimated values from the replaced unit).

5 - Replacing the carbon discs before reaching wear threshold (“Disc Wear Index” parameter < 100%)

This is a non-standard procedure and may only be performed by authorised personnel. Contact the Ferrari SAT service.